

2022.9.14
wed
BNU

微积分

§ 1.1 映射与函数.

~~定义~~ map.

2022.9.16
Fri

1. 映射概念

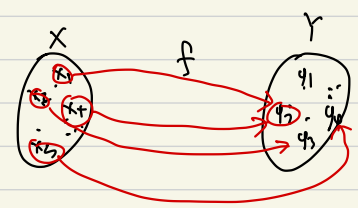
定义 设 X, Y 是两个非空集合, 如果存在一个法则 f , 使得对 X 中每个元素 x , 按法则 f , 在 Y 中有唯一确定的元素 y 与之对应, 那么称 f 为从 X 到 Y 的映射, 记作

$$f: X \rightarrow Y,$$

其中 y 称为元素 x (在映射 f 下) 的像, 并记作 $f(x)$, 即

$$y = f(x),$$

而元素 x 称为元素 y (在映射 f 下) 的一个原像; 集合 X 称为映射 f 的定义域, 记作 D_f , 即 $D_f = X$; X 中所有元素的像所组成的集合称为映射 f 的值域, 记作 R_f 或 $f(X)$, 即

$$R_f = f(X) = \{f(x) \mid x \in X\}.$$


原像. 象.

- ① 单射. 原像是唯一的.
- ② 满射. Y 中每个元素都有原像.
- ③ 双射, -- 映射.

集合论

$\{ \{1, x_1\}, \{2, y_2\} \}$
 $\{x_1, \{y_2\}\}$

$f: \left\{ \begin{array}{l} \{x_1, y_2\} \\ \{x_2, y_3\} \\ \dots \end{array} \right\}$

~~多值映射~~

约定: 映射是单值映射的简称.

思考

- 点, 直线 如何定义的?
- 公理化 ~~X~~
- $f(x; \lambda) = (1-\lambda)x + \lambda x^2$
- 所有集合的集合?
- $0 := \emptyset$
- $1 := \{\emptyset\}$
- $2 := \{0, 1\} = \{\emptyset, \{\emptyset\}\}$
- $\vdots$

映射例子

• 函数

$$f(x) = \sin x.$$

$$\mathbb{R} \rightarrow \mathbb{R}$$

X
O

$\mathbb{R}$

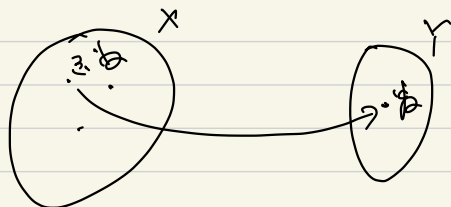
$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Y
O
[-1,1]

$$Y = \mathbb{R}$$

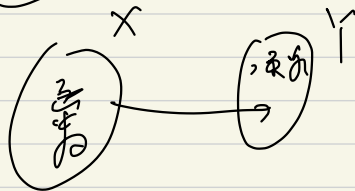
$$f(x, y) = x + y$$

• 函数



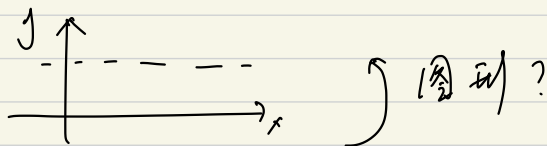
$$S(t) \rightarrow E[S(t)]$$

• 变换



$$S(t) \rightarrow v(t)$$

函数 $\rightarrow$ 数对集合.



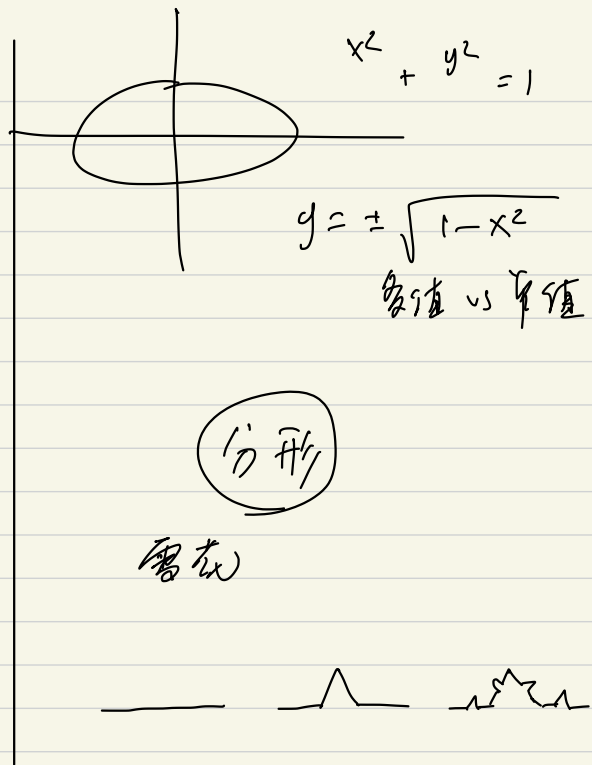
eg: Dirichlet 函数

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Riemann 函数

$$R(x) = \begin{cases} \frac{1}{p}, & x = \frac{q}{p} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

$|x|$, $\text{sgn}(x)$, $[x]$, ...



制造更多的函数

四则运算

复合, 反函数

初等函数

u 为整数, 可按单项式, 分来理解
 u 不是整数时,
 $\ln(x^u)$

$$= e$$

$$x^u = e^{u \ln x}, (x > 0)$$

$$x^u, u \in \mathbb{Z}$$

$$x^2, x^3, \dots$$

$$\frac{1}{x^2}, \frac{1}{x^3}, \dots$$

$$x^\pi, x^{\frac{22}{7}}$$

$$\left(x^{22} \right)^{\frac{1}{7}}$$

$$x^{\frac{22}{7}}$$

缺点: $x^3 = x^{\frac{6}{2}} \neq (x^6)^{\frac{1}{2}}$

在初等数学中已经讲过下面几类函数:

幂函数 ($y = x^\mu$ ($\mu \in \mathbb{R}$ 是常数)),

指数函数: $y = a^x$ ($a > 0$ 且 $a \neq 1$),

对数函数: $y = \log_a x$ ($a > 0$ 且 $a \neq 1$, 特别当 $a = e$ ① 时, 记为 $y = \ln x$),

三角函数: 如 $y = \sin x, y = \cos x, y = \tan x$ 等,

反三角函数: 如 $y = \arcsin x, y = \arccos x, y = \arctan x$ 等.

以上这五类函数统称为基本初等函数.

由常数和基本初等函数经过有限次的四则运算和有限次的函数复合步骤所构成并可用一个式子表示的函数, 称为初等函数. 例如

$$y = \sqrt{1-x^2}, \quad y = \sin^2 x, \quad y = \sqrt{\cot \frac{x}{2}}$$

等都是初等函数. 在本课程中所讨论的函数绝大多数都是初等函数.

$|x|$ 是不是初等函数? $\checkmark$

$$= \sqrt{x^2}$$



$$\text{eg: } \sin\left(\frac{x-|x|}{2}\right) + \left(\frac{x+|x|}{2}\right)^2 = \begin{cases} \sin x, & x < 0 \\ x^2, & x > 0 \end{cases}$$

函数的性质.

- 单: 周
 - 奇偶
 - 周期
 - 对称.
 - 四: 品
 - 有界.
- 导数

• $\sin x + \sin(\pi x)$, 不是周期
文. 为.

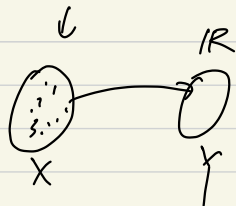
$$\sin x + \sin y.$$

$$y = \pi x.$$

注意

2022.9.19
Mon

§ 1.2 数列的极限



$\{a_n\}$, 数列 a_n .

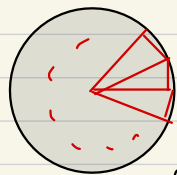
Fibonacci: $a_{n+1} = a_n + a_{n-1}$, $a_0 = 0$, $a_1 = 1$, $a_2 = 1$, 2, 3, 5, 8, ...

$$a_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right) \approx \lambda \varphi^n$$

if $a_n = \lambda_1 \varphi_1^n + \lambda_2 \varphi_2^n$

$\lambda \varphi^n$
| 差分方程

$$\sqrt[n]{a_n} \rightarrow \frac{1+\sqrt{5}}{2}, \quad n \rightarrow \infty$$



割圆术

$A_n = 2n \sin \frac{\pi}{n} \rightarrow 2\pi$

$\triangle$ $\theta = \frac{2\pi}{n}$

$\{\varphi^n\}$
 $\varphi = \frac{1}{2}$
 $\frac{1}{2^n} \rightarrow 0$

$$a_{n+1} = \frac{1}{1+a_n}$$

连分数

$$= \frac{1}{1 + \frac{1}{1+a_{n-1}}}$$



连分数与迭代

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}}$$

0.618... ✗

$$\frac{\sqrt{5}-1}{2}$$



$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) \rightarrow \sqrt{a}$$

$x_0 > 0$ (求根)

$$x = \frac{1}{1+x}$$

$$x^2 + x - 1 = 0$$

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \ln n \rightarrow \gamma$$

Euler 常数

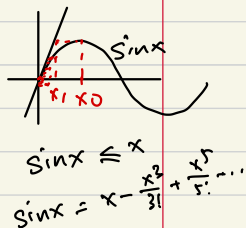
$$x_1 \in (0, 1)$$

$$x_{n+1} = \sin(x_n) \rightarrow 0$$

$$x_{n+1} = \cos(x_n) \rightarrow 0.73 \dots$$

$$x = \cos x$$

$$x_{n+1} = 4x_n(1-x_n), \quad \text{混沌}, \text{ Logistic 映射}$$



a_n 逐步接近某个数 a .

$$\exists a \in \mathbb{R}, \text{ s.t.}$$

$$\forall \varepsilon, \exists N, \text{ s.t.}$$

$$|a_n - a| < \varepsilon, \forall n > N$$

① 定义: $\{a_n\}$ 收敛于 a .

$\{a_n\}$ 收敛

" ε - N " 语言:

$\forall \varepsilon > 0$
Arbitrary
任意

$\exists N \in \mathbb{Z}^+, \text{ s.t.}$
Exist
to be.
Such that
使得

$$|a_n - a| < \varepsilon, \forall n > N$$

$$a - \varepsilon < a_n < a + \varepsilon$$

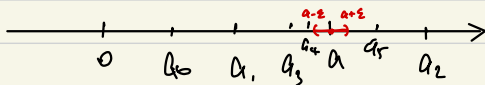
$$|a_n - a| \text{ "足够小"}$$

充分小.

$$\forall \varepsilon > 0, \exists N, \text{ s.t.}$$

$$|a_n - a| < \varepsilon, \forall n > N$$

$$\lim_{n \rightarrow \infty} a_n = a$$



$$\forall \varepsilon > 0, \exists N \in \mathbb{Z}^+, \text{ s.t.}$$

$$|a_n - a_m| < \varepsilon, \forall m, n > N$$

$\{a_n\}$ 趋于 a

$\{a_n\}$ 以 a 为极限.

$\{a_n\}$ 不收敛于 a . $\neg (\lim_{n \rightarrow \infty} a_n = a)$

$$\exists \varepsilon > 0, \text{ s.t. } \forall N \in \mathbb{Z}^+, \exists n > N, |a_n - a| \geq \varepsilon$$

$\{\frac{1}{2^n}\}$ 收敛于 0. 指的是:

命题 vs 证明
定理.

$$\forall \varepsilon > 0, \boxed{\exists} N = \lfloor \log_2(\frac{1}{\varepsilon}) \rfloor + 1 \in \mathbb{N}.$$

$$|\frac{1}{2^n} - 0| < \varepsilon, \quad \forall n > N.$$

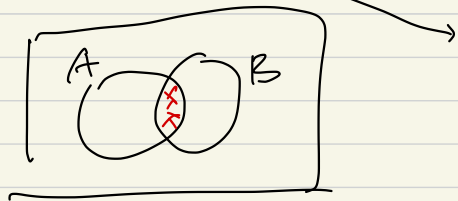
$$\frac{1}{2^n} < \varepsilon, \quad 2^n > \frac{1}{\varepsilon}, \quad n > \underline{\log_2(\frac{1}{\varepsilon})}$$

$\{\frac{1}{2^n}\}$ 收敛于 1. $\leftarrow$ 假命题.

2022.8.21

$$x \notin A \cap B \iff x \notin A \text{ or } x \notin B.$$

逻辑运算



$$\begin{aligned} & \neg(x \in A \cap B) \\ & \neg(x \in A \wedge x \in B) \\ & \neg(x \in A) \vee \neg(x \in B) \end{aligned}$$

3.4 牛顿法应用举例

对于给定正数 C , 应用牛顿法解二次方程 $x^2 - C = 0$, 可导出求开方值 $\sqrt{C}$ 的计算程序

$$x_{k+1} = \frac{1}{2} \left(x_k + \frac{C}{x_k} \right). \quad (3.7)$$

我们现在证明, 这种迭代公式对于任意初值 $x_0 > 0$ 都是收敛的. 事实上, 对 (3.7) 式施行配方手续, 易知

$$x_{k+1} - \sqrt{C} = \frac{1}{2x_k} (x_k - \sqrt{C})^2; \quad x_{k+1} + \sqrt{C} = \frac{1}{2x_k} (x_k + \sqrt{C})^2.$$

以上两式相除得 $\frac{x_{k+1} - \sqrt{C}}{x_{k+1} + \sqrt{C}} = \left[\frac{x_k - \sqrt{C}}{x_k + \sqrt{C}} \right]^2$. 据此反复递推有

$$\frac{x_k - \sqrt{C}}{x_k + \sqrt{C}} = \left[\frac{x_0 - \sqrt{C}}{x_0 + \sqrt{C}} \right]^{2^k}, \quad (3.8)$$

记 $q = \frac{x_0 - \sqrt{C}}{x_0 + \sqrt{C}}$, 整理 (3.8) 式, 得 $x_k - \sqrt{C} = 2\sqrt{C} \frac{q^{2^k}}{1 - q^{2^k}}$.

对任意 $x_0 > 0$, 总有 $|q| < 1$, 故由上式推知, 当 $k \rightarrow \infty$ 时 $x_k \rightarrow \sqrt{C}$, 即迭代过程恒收敛.

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) \approx \sqrt{a}, \quad x_0 > 0$$

$$x_{n+1} - \sqrt{a} = \frac{1}{2x_n} (x_n - \sqrt{a})^2, \quad n \geq 1$$

$$\leq \frac{1}{2} \frac{x_n - \sqrt{a}}{x_n} (x_n - \sqrt{a})$$

$$0 \leq 1 - \frac{\sqrt{a}}{x_n} \leq 1$$

$$\leq \frac{1}{2} (x_n - \sqrt{a})$$

$$\leq \dots \leq \frac{1}{2^{n+1}} (x_0 - \sqrt{a})$$

$\rightarrow 0.$

$$x_{n+1} = \sin(x_n) \rightarrow 0$$

证明思路 1°: 不妨设 $x_0 \in (0, 1)$, 则 $\{x_n\}$ 单调有界, 有极限 a . 且 $a = \sin(a)$
只可能 $a = 0$.

2°. 若利用定义来证, 由于 $x_n \in (0, 1)$,

$$\text{利用不等式 } \sin x \leq x - \frac{x^3}{8}, \quad x \in (0, 1)$$

$$\text{则 } \sin^2 x \leq x^2 - \frac{x^4}{4} + \frac{x^6}{64} \leq x^2 - \frac{x^4}{8}$$

$$\frac{8}{x_{n+1}^2} = \frac{8}{\sin^2 x_n} \geq \frac{1}{\frac{1}{8}x_n^2(1-\frac{x_n^2}{8})} = \frac{1}{\frac{1}{8}x_n^2} + \frac{1}{1-\frac{1}{8}x_n^2} \geq \frac{8}{x_n^2} + 1$$

$$\geq \dots \geq \frac{8}{x_1^2} + n \geq n+1$$

从而 $x_n \leq \frac{8}{\sqrt{n}} < \varepsilon$

对 $\forall \varepsilon > 0$. 取 $N = \lfloor \frac{8}{\varepsilon} \rfloor^2$, 则 $|x_n - 0| < \varepsilon, \forall n > N$

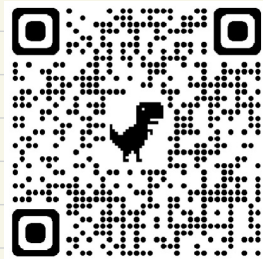
★ 证明 $a_{n+1} = a_n(1-a_n)$, $a_0 \in (0,1)$, 收敛于 0.

思路: $\frac{1}{a_{n+1}} = \frac{1}{a_n} + \frac{1}{1-a_n} \geq \frac{1}{a_n} + 1 \geq \dots \geq n+1$

从而 $a_n \leq \frac{1}{\sqrt{n}} \rightarrow 0$.

注: 证明数列收敛, 以及求数列的极限
非常有技巧性

这里的例子仅供参考, 不作为考试要求.



2022.9.21
Wed

§ 1.2 数列的极限

$$\lim_{n \rightarrow \infty} a_n = a$$

$$\varepsilon = \frac{b-a}{4}$$

② 性质

$$\lim_{n \rightarrow \infty} a_n = a, \lim_{n \rightarrow \infty} a_n = b \Rightarrow a = b.$$

1°. 唯一性

2°. 有界性 $\rightarrow \exists M > 0, \text{ s.t. } |a_n| < M, \forall n$

* 3°. 保号性 $\rightarrow$ 若 $a_n > 0$ 则 $a \geq 0$.

反证
设 $a < 0$

* 4°. 子列收敛性 $\rightarrow$ 子列 $\{a_{n_k}\}$ 也收敛于 a

5°. 四则运算

* 若 $\lim_{n \rightarrow \infty} a_n = a > 0$

$\Rightarrow \exists N, \text{ s.t. } a_n \geq \frac{a}{2} > 0, \forall n > N.$

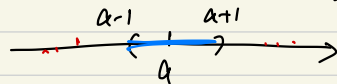
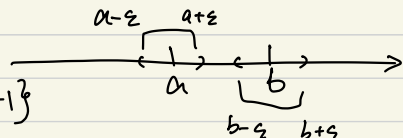
* 某个 $\{a_{n_k}\}$ 收敛 $\nRightarrow \{a_n\}$ 收敛

eg. $(-1)^n$

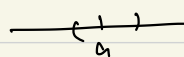
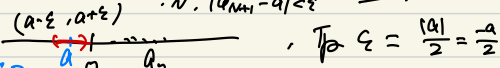
* $\{a_{2k}\} \rightarrow a, \{a_{2k+1}\} \rightarrow a$
 $\Rightarrow \{a_n\} \rightarrow a.$

④ 单调有界 $\rightarrow$ 收敛.

$$M \text{ 取为 } \max\{|a_1|, \dots, |a_n|, |a+1|\}$$



$$n, |a_{n+1} - a| < \varepsilon \Rightarrow a_{n+1} < a + \frac{|a|}{2} = \frac{a}{2} < 0$$



$$\left. \begin{aligned} \lim_{n \rightarrow \infty} a_n &= a \\ \lim_{n \rightarrow \infty} b_n &= b \end{aligned} \right\} \Rightarrow \lim_{n \rightarrow \infty} (a_n + b_n) = a + b \quad n > N_1$$

$$\begin{aligned} |a_n - a| &\leq \frac{\varepsilon}{2} \\ |b_n - b| &\leq \frac{\varepsilon}{2} \\ |a_n + b_n - (a + b)| &\leq |a_n - a| + |b_n - b| \leq \varepsilon, \quad n > N_1 + N_2 \end{aligned}$$

$$\lim_{n \rightarrow \infty} (a_n b_n) = ab$$

$$\forall \varepsilon > 0, \exists N = \max(N_1, N_2), \text{ 当 } n > N \text{ 时 } |b_n - b| \leq \frac{\varepsilon}{2M}, \quad n > N_1$$

$$|a_n b_n - ab| < \varepsilon.$$

$$\leq |a_n| |b_n - b| + |b| |a_n - a| \leq \varepsilon$$

证明: $a_n = (-1)^n$ 不收敛.

①. $\exists \varepsilon = \frac{1}{2}$, s.t. $\forall N > 0$, $\exists n = N$, $\exists n = N+1$, $(a_n - a_{n+1}) = 2 > \varepsilon$

② 若 $\{a_n\} \rightarrow a$, $\{a_{2k}\} \rightarrow a$, $\{a_{2k+1}\} \rightarrow a$; $l = -1$, 矛盾.

③ $\forall a$, $\{a_n\} \not\rightarrow a$

证: $\{a_{2k}\} \rightarrow a$, $\{a_{2k+1}\} \rightarrow a$ 则 $\{a_n\} \rightarrow a$.

$\lim_{n \rightarrow \infty} a_n = a$, $\lim_{n \rightarrow \infty} b_n = b$, $b_n \neq 0$, $b \neq 0$
不取 $b > 0$

$\Rightarrow \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}$

只需证 $\lim_{n \rightarrow \infty} \frac{1}{b_n} = \frac{1}{b}$

利用保号性 $b_n \geq \frac{b}{2} > 0$, $n > N$

$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{|b_n - b|}{|b_n b|} \leq \frac{2}{|b|^2} |b_n - b| < \varepsilon$

$\Uparrow |b_n - b| < \frac{|b|^2 \varepsilon}{2}$

2022.9.23

Fri

§1.3 函数的极限

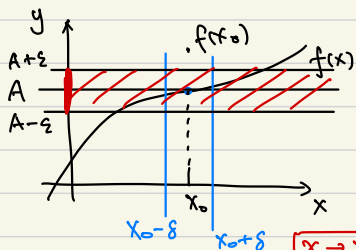
$x, \frac{x^2}{x}$, 定义域.

① 定义. $f(x)$ 在 x_0 处的极限. 为 A

" ϵ - δ " 语言.

$$\lim_{x \rightarrow x_0} f(x) = A$$

$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } |f(x) - A| < \epsilon, \forall x, 0 < |x - x_0| < \delta$

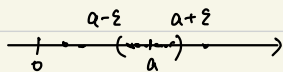
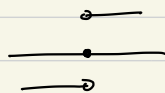


$$\forall x \in (x_0 - \delta, x_0 + \delta)$$

$$\forall x \in U^0(x_0, \delta)$$

$$(x_0 - \delta, x_0) \cup (x_0, x_0 + \delta)$$

sgn(x)



例: $\lim_{x \rightarrow x_0} \sqrt{x} = \sqrt{x_0}, \quad x_0 > 0.$

$$x \neq x_0$$

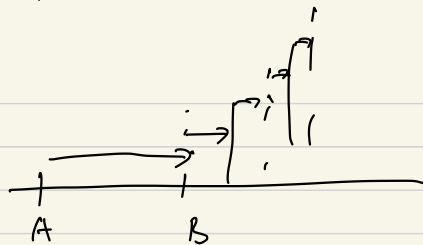
$$x_0 - \delta < x < x_0 + \delta$$

$\forall \epsilon > 0, \exists \delta = \min\left(\epsilon \sqrt{x_0}, \frac{x_0}{2}\right), \text{ s.t. } 0 < |x - x_0| < \delta$

$$|\sqrt{x} - \sqrt{x_0}| < \epsilon, \quad \forall x, 0 < |x - x_0| < \delta.$$

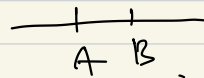
$$= \frac{|x - x_0|}{\sqrt{x} + \sqrt{x_0}} \leq \frac{1}{\sqrt{x_0}} |x - x_0| < \frac{\delta}{\sqrt{x_0}} < \epsilon, \quad \delta < \epsilon \sqrt{x_0}$$

$$0 + 1 + 2 + 3 + 4 + \dots = -\frac{1}{12}$$



"无界"

①. ②



"一样多" = "

"一对应"

$$a \in (0, 1),$$

$$a_1 = 0. a_1^{(1)} a_1^{(2)} \dots$$

$$a_2 = 0. a_2^{(1)} a_2^{(2)} \dots$$

$$\vdots$$

$$a_n = 0. a_n^{(1)} a_n^{(2)} \dots$$

$$\vdots$$

$$b = 0. b_1 b_2 \dots$$

$$b_i \neq a_i^{(1)}, \dots, b_n \neq a_n^{(n)}$$

有理数
可数(列)
无理数
不可数
实数
不可数

$$\textcircled{x} N_0, N_1 = 2^{N_0}, \dots$$

$$① \lim_{x \rightarrow x_0} f(x) = A$$

$$② \lim_{x \rightarrow x_0^+} f(x) = A \quad \text{右}$$

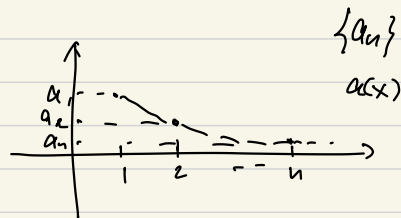
$$③ \lim_{x \rightarrow x_0^-} f(x) = A \quad \text{左}$$

$$\lim_{x \rightarrow 0^+} \operatorname{sgn}(x) = 1$$

$$\lim_{x \rightarrow 0^-} \operatorname{sgn}(x) = -1$$

$$\operatorname{sgn}(x) = \begin{cases} +1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

$$\lim_{x \rightarrow 0} \operatorname{sgn}(x) \text{ 不存在}$$



$$① \Rightarrow ② \Rightarrow ③$$

条件: 左右极限存在且相同

$$④ \lim_{x \rightarrow \infty} f(x) = A.$$

$$\forall \varepsilon > 0, \exists N, \text{ s.t. } |f(x) - A| < \varepsilon, \forall |x| > N.$$

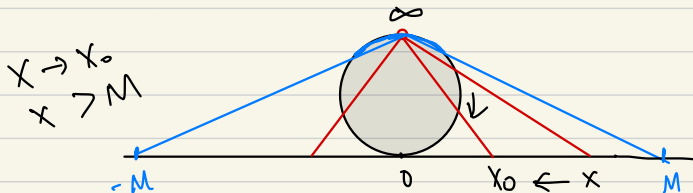
$$\lim_{x \rightarrow +\infty} f(x) = A$$

$$\lim_{x \rightarrow -\infty} f(x) = A.$$

$$\text{eg: } \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

注: $x_0 \neq \infty$

$$\lim_{x \rightarrow x_0^+} = \lim_{x \rightarrow +\infty}$$



2022.9.26
Mon

② 性质

$$\lim_{x \rightarrow x_0} f(x) = A$$

1° 唯一性 $\lim_{x \rightarrow x_0} f(x) = a, \lim_{x \rightarrow x_0} f(x) = b \Rightarrow a = b.$

2° 局部有界性, $(\frac{1}{x}, x_0 \neq 0)$, $\exists \delta > 0$, 使 $f(x)$ 在 $(x_0 - \delta, x_0) \cup (x_0, x_0 + \delta)$ 上有界.

3° 局部保号性 $\left\{ \begin{array}{l} \text{若 } A > 0, \text{ 则 } \exists \delta > 0, \text{ 使 } f(x) \geq \frac{A}{2} > 0. \quad x \in (x_0 - \delta, x_0 + \delta) \setminus \{x_0\} \\ f(x) > 0 \Rightarrow A \geq 0 \end{array} \right.$

4° ~~判定~~ 恒等原则. $\lim_{x \rightarrow x_0} f(x) = A \iff \forall \epsilon > 0, \text{ 若 } \{x_n\} \rightarrow x_0, \text{ 则 } \{f(x_n)\} \rightarrow A.$
 $x_n \neq x_0$

5° 四则运算. $\begin{array}{l} x(t) \rightarrow x_0 \\ x_n = x(t_n) \end{array}$

$$\lim_{x \rightarrow x_0} f(x) = A, \lim_{x \rightarrow x_0} g(x) = B$$

$$\Rightarrow \lim_{x \rightarrow x_0} (k_1 f(x) \pm k_2 g(x)) = k_1 A \pm k_2 B$$

$$\lim_{x \rightarrow x_0} (f(x) g(x)) = AB = \left(\lim_{x \rightarrow x_0} f(x) \right) \left(\lim_{x \rightarrow x_0} g(x) \right)$$

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{A}{B} \quad (g(x) \neq 0, B \neq 0)$$

四则运算与取极限
可以交换顺序.

例:

$$\lim_{x \rightarrow x_0} (f(x)g(x)) = AB = \left(\lim_{x \rightarrow x_0} f(x) \right) \left(\lim_{x \rightarrow x_0} g(x) \right)$$

注: 这是利用 $\lim_{x \rightarrow x_0} f(x) = A$
 $\forall \varepsilon_1 > 0, \exists \delta > 0, \text{ s.t. } |f(x) - A| < \varepsilon_1, x \in \dots$
这里取 $\varepsilon_1 = \frac{\varepsilon}{2|B|+1}$

证明: 由于 $\lim_{x \rightarrow x_0} f(x) = A, \lim_{x \rightarrow x_0} g(x) = B,$

$\varepsilon \in (0, 1+2|B|)$

利用定义, $\forall \varepsilon \in (0, 1)$, $\exists \delta > 0$, s.t.

$$|f(x) - A| < \min\left(1, \frac{\varepsilon}{2|B|+1}\right)$$

$$|g(x) - B| < \frac{\varepsilon}{2(|A|+1)}, \quad x \in (x_0 - \delta, x_0 + \delta) \setminus \{x_0\}$$

注: 强调的充分小.

$$|f(x)| < |A|+1$$

$$\text{则 } |f(x)g(x) - AB| = |f(x)g(x) - f(x)B + f(x)B - AB|$$

$$\leq |f(x)| |g(x) - B| + |f(x) - A| |B|$$

$$\leq (|A|+1) \frac{\varepsilon}{2(|A|+1)} + \frac{\varepsilon}{2|B|+1} \cdot |B|$$

$$\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

因此 $\lim_{x \rightarrow x_0} f(x)g(x) = AB$

2022.9.26

Mon

§ 1.4 无穷小与无穷大

定义: 无穷小 — (当...时) 是无穷小.

不是零, 而是趋势.

若 $\lim_{x \rightarrow x_0} f(x) = 0$ 则称 $f(x)$ 当 $x \rightarrow x_0$ 时为无穷小.

无穷大

$$\lim_{x \rightarrow x_0} f(x) = \infty.$$

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } |f(x) - A| < \varepsilon, \forall 0 < |x - x_0| < \delta.$$

$$f(x) \in (A - \varepsilon, A + \varepsilon), \quad x \in (x_0 - \delta, x_0 + \delta) \\ x \neq x_0$$

$$|f(x)| > M$$

$$|x| > N$$

$$\lim_{x \rightarrow \infty} f(x) = \infty.$$

	$f \rightarrow A$	$f \rightarrow \pm\infty$
$x \rightarrow x_0^\pm$		
$x \rightarrow \pm\infty$		

$$\lim_{x \rightarrow x_0} f(x) = A$$

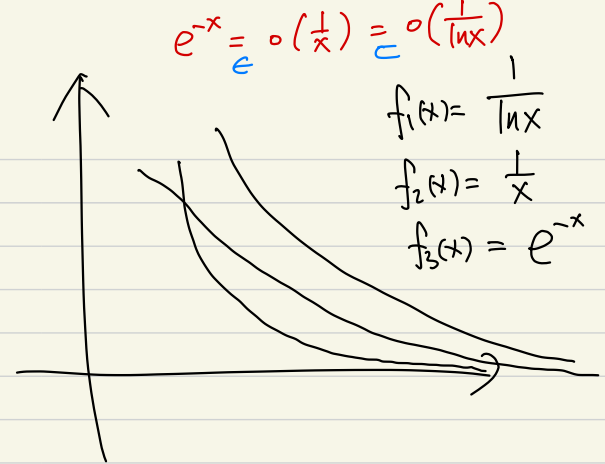
$$\forall M > 0, \exists N > 0, \text{ s.t. } |f(x)| > M, \forall |x| > N.$$



§1.7 无穷小(大)的比较

$f(x) \rightarrow 0, \quad g(x) \rightarrow 0.$

考察 $\boxed{\frac{f(x)}{g(x)}}$



$\lim \frac{f(x)}{g(x)} = \begin{cases} 0, & \text{于是 } g \text{ 的 } \textcircled{\text{高}} \text{ 阶无穷小.} \\ c \ (c \neq 0), & \textcircled{\text{同阶}} \\ \infty, & \text{--- } \textcircled{\text{低}} \text{ 阶无穷小.} \\ \text{不存在.} & \text{不可比较} \end{cases} \quad f = o(g)$

特别 $c=1$, $\textcircled{\text{等价}}$ 无穷小.

$f \sim g$

$\lim \frac{f(x)}{g^k(x)} = c \in (0, \infty), \quad \dots \dots k \text{ 阶无穷小.}$
 $(k > 0) \quad f \sim c g^k(x)$

$f = O(g), \quad |\frac{f}{g}| \text{ 有界. } |f| \leq M|g|$

- 例: 1. $f \sim f$
 2. $f \sim g \Rightarrow g \sim f$
 3. $f \sim g, g \sim h \Rightarrow f \sim h$

注: $f = o(g) \Rightarrow f = O(g)$, $o(g) \subset O(g)$

$\left. \begin{matrix} f_1 = o(g) \\ f_2 = o(g) \end{matrix} \right\} \nRightarrow f_1 = f_2$ → 应当理解成 $f \in o(g)$

例: $g(x) = x$. $\begin{cases} f(x) = x^2, & f = o(g), & x \rightarrow 0 \\ f(x) = \sin x & f \sim g \\ f(x) = \sqrt{|x|} & \text{低阶} \end{cases}$, $\lim_{x \rightarrow 0} \frac{\sqrt{|x|}}{x} = \lim_{x \rightarrow 0} \frac{1}{\sin(x)\sqrt{|x|}} \rightarrow \infty$

小生版

应用 $\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow f(x) = A + o(1)$

★ $\alpha \sim \tilde{\alpha}, \beta \sim \tilde{\beta} \Rightarrow \lim \frac{\beta}{\alpha} = \lim \frac{\tilde{\beta}}{\tilde{\alpha}}$ (若存在)

(有用)

2022.9.28
wed

§ 1.5 极限的运算法则

$$\lim_{x \rightarrow x_0} f(x) = A, \quad \lim_{x \rightarrow x_0} g(x) = B$$

$$\textcircled{1} \Rightarrow \begin{array}{ccc} f \pm g & , & f \cdot g & , & f/g \quad (g \neq 0, B \neq 0) & \text{极限存在.} \\ \downarrow & & \downarrow & & \downarrow & \\ A \pm B & & AB & & A/B & \end{array}$$

$$\textcircled{2} \left. \begin{array}{l} \lim f \text{ 存在.} \\ \lim g \text{ 不存在} \end{array} \right\} \Rightarrow \begin{array}{l} \lim (f \pm g) \text{ 不存在.} \\ \lim \left. \begin{array}{l} fg \\ f/g \end{array} \right\} \text{ 有可引存在.} \end{array}$$

eg. $f(x) = x, \quad g(x) = \frac{1}{x}, \quad x_0 = 0. \quad (fg)(x) = 1 \quad \checkmark.$

$$g(x) = \sin\left(\frac{1}{x}\right), \quad x_0 = 0. \quad (fg)(x) \rightarrow 0$$

✓ ⊛ 若 $\lim_{x \rightarrow x_0} f(x) = 0$, 而且 $g(x)$ 有界. 则 $\lim_{x \rightarrow x_0} (fg)(x) = 0$
(f 是 $x \rightarrow x_0$ 时的无穷小)

$$|fg - 0| \leq \underbrace{|f|}_{\leq \frac{\varepsilon}{M}} \underbrace{|g|}_{\leq M} \leq \varepsilon$$

约定.

“收敛”,

极限 $\in \mathbb{R}$.

eg. $f(x) = \sin x$, $g(x) = \tan x$, $x_0 = \frac{\pi}{2}$. $f(x)/g(x) = \cos x \rightarrow 0$.

✓ ⊗ 若 $\lim_{x \rightarrow x_0} f(x)$ 存在, $g(x)$ 为无穷大, 则 $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0$.
 { 或 $f(x)$ 有界.

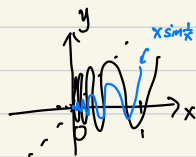
✓ ⊗ $\uparrow$
 $g(x)$ 为无穷大 $\Rightarrow \frac{1}{g(x)}$ 为无穷小. ($g(x) \neq 0$)
 $\Leftarrow$

⊗ $f(x)$ 为无穷小 $\Rightarrow \frac{1}{f(x)}$ 为无穷大, ($f(x) \neq 0$)

推论: Page 38, 39, 41.

Ⓛ + - x ÷ 幂. 多项式, 有理函数

eg. $\lim_{x \rightarrow x_0} f(x) = A$, $\Rightarrow \lim_{x \rightarrow x_0} \frac{f^5(x) - f^3(x) + 1}{f^2(x) + 1} = \frac{A^5 - A^3 + 1}{A^2 + 1}$



eg. $f(x) = x \cdot \sin(\frac{1}{x})$
 $\frac{1}{f(x)} = \frac{1}{x} \frac{1}{\sin(\frac{1}{x})}$

eg. $g(x) = x \cos x$.
 $x \rightarrow \infty$

函数复合. $\leftarrow$ 构造更复杂的函数.

$$\lim_{y \rightarrow y_0} f(y) = A. \quad \lim_{x \rightarrow x_0} \boxed{g(x)} = B = y_0$$

$$\Rightarrow \lim_{x \rightarrow x_0} f(\boxed{g(x)}) = A$$

条件: $g(x) \neq y_0, x \in U^\circ(x_0; \delta).$

证明: $\forall \varepsilon > 0, \exists \delta = \underline{\hspace{2cm}} \delta_2$

$$|f(y) - A| < \varepsilon, \quad \boxed{y \in U^\circ(y_0; \delta_1)}$$

$$|y - y_0| = |g(x) - y_0| < \delta_1, \quad g(x) \neq y_0$$

$$\Uparrow \forall x \in U^\circ(x_0; \delta_2)$$

$$|f(g(x)) - A| < \varepsilon, \quad x \in U^\circ(x_0; \delta).$$

$$\begin{matrix} \text{"} \\ |f(y) - A| \end{matrix}$$

f. g

$$(f \circ g)(x) = f(g(x))$$

$$(g \circ f)(x) = g(f(x))$$

$$\begin{matrix} x \rightarrow x_0, & g(x) \rightarrow B \\ y \rightarrow \boxed{B} \\ y \rightarrow y_0, & f(y) \rightarrow A \end{matrix}$$

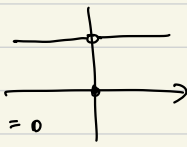
$y \neq y_0$

例: $f(y) = |\operatorname{sgn}(y)|, y_0 = 0$

$$\lim_{y \rightarrow 0} f(y) = 1$$

$$g(x) = 0, x_0 = 0, g(x) \rightarrow 0.$$

$$f(g(x)) = 0 \rightarrow 0$$



§ 1.6 极限存在准则

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 0, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\begin{cases} \lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x = e \approx 2.718... \\ \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e \end{cases}, \text{自然对数的底数.}$$

$$\lim_{n \rightarrow \infty} (1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n) = \gamma \approx 0.577... \text{欧拉常数.}$$

- ① 存在
- ② 求值.

$$r_n = \frac{r}{n}$$

$$1+r$$

$$(1+\frac{r}{3})^3 = 1 + 3 \cdot \frac{r}{3} + \dots$$

$$(1+\frac{r}{3^{1/2}})^{3^{1/2}}$$

例: $\frac{\sin x}{x} \rightarrow 1 \quad (x \rightarrow 0)$ page 46.

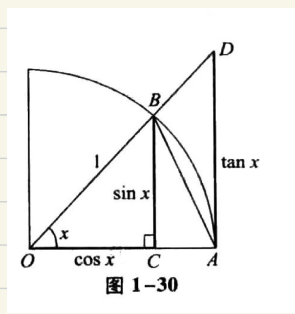
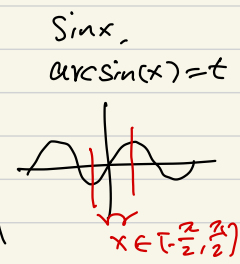
$$\boxed{\cos x} \leq \frac{\sin x}{x} \leq \boxed{1}$$

$x \leq \tan x$ $\sin x \leq x$

证: $\lim_{x \rightarrow 0} \cos x = 1$.

$$|\cos x - 1| = 2 \sin^2 \frac{x}{2}$$

$$\leq 2 (\frac{x}{2})^2 = \frac{x^2}{2} < \varepsilon.$$



$$\sin x \leq x \leq \tan x$$

证: $\lim_{x \rightarrow 0} \cos x = 1$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\arcsin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\arctan x}{x} = 1.$$

★ 收敛准则. (第21)

① 夹逼准则. $\{x_n\}$ (证明 page 46)

$$y_n \leq x_n \leq z_n, \quad y_n \rightarrow a, \quad z_n \rightarrow a. \Rightarrow x_n \rightarrow a.$$

② 单调有界准则. $\{x_n\}$

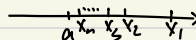
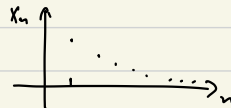
$$x_{n+1} \leq x_n, \text{ 并且 } x_n \geq m. \Rightarrow \{x_n\} \text{ 收敛.}$$

单调递减且有下界

证明: 利用实数的完备性质

↳ 有界必有 确界

下确界记为 a . 则 $x_n \rightarrow a$



$\forall \epsilon > 0, a + \epsilon$ 不是下界. $|x_n - a| < \epsilon$
 $a < x_n < a + \epsilon, n > N.$

③ 柯西 (Cauchy) 准则. $\{x_n\}$

$$\forall \epsilon > 0, \exists N > 0, \text{ s.t. } |x_n - x_m| < \epsilon, \quad \forall m, n > N.$$

证明: $\inf_{k \geq n} \{x_k\} \leq x_n \leq \sup_{k \geq n} \{x_k\}$
 记 $y_n = \inf_{k \geq n} \{x_k\}$ (下确界)
 记 $z_n = \sup_{k \geq n} \{x_k\}$ (上确界)
 显然 $y_n \leq x_n \leq z_n$ 且 $y_n \rightarrow a_-, z_n \rightarrow a_+$

若 $a_- < a_+$ 取 $\epsilon = \frac{a_+ - a_-}{4}$

存在 $n, y_n = \inf_{k \geq n} \{x_k\} \in (a_-, a_+)$

进而有某 k_1 使

$$x_{k_1} \in (a_-, a_+)$$

同理有某 k_2 使

$$x_{k_2} \in (a_-, a_+)$$

则 $|x_{k_1} - x_{k_2}| > a_+ - \epsilon - (a_- + \epsilon) > \epsilon$
 矛盾

思考: 思考.

2022.10.3

$$\oplus \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

二项展开

$$x_n = \left(1 + \frac{1}{n}\right)^n = 1 + n \cdot \frac{1}{n} + \frac{n(n-1)}{2} \cdot \left(\frac{1}{n}\right)^2 + \dots + \frac{n(n-1)\dots 1}{n!} \left(\frac{1}{n}\right)^n$$

$$= 1 + 1 + \frac{1}{2} \frac{n}{n} \frac{n-1}{n} + \dots$$

$$= 1 + 1 + \frac{1}{2} \underbrace{1 \cdot \left(1 - \frac{1}{n}\right)} + \frac{1}{3!} \underbrace{\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right)} + \dots$$

每一项都递增.

page 48.

有界

x_n 有界



$$x_n \leq 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!}$$

$$\leq 1 + 1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{n-1}}$$

$$\leq 1 + \frac{1}{1 - \frac{1}{2}} = 3$$

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$$

$$\uparrow \quad \downarrow$$

$$\geq 1 \cdot 2 \cdot 2 \cdot \dots \cdot 2$$

故 x_n 收敛. 其极限记为 e .

不等式：
夹逼准则

$$\textcircled{+} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$x \rightarrow +\infty \Rightarrow$$

$$x \rightarrow -\infty$$

$$\boxed{x = -t}$$

$$\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x = \frac{1}{e}$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$$

$$\boxed{\lim_{n \rightarrow \infty} \left(1 \pm \frac{1}{n}\right)^n = e^{\pm 1}}$$

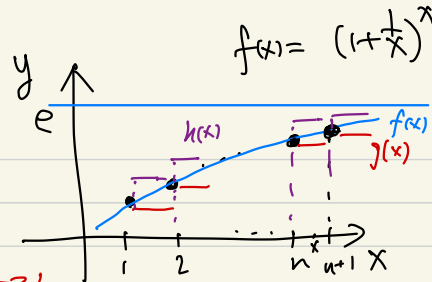
技巧：
倒数法

$$\left(1 + \frac{1}{x}\right)^x = \left(1 - \frac{1}{t}\right)^{-t} = \frac{1}{\left(\frac{t-1}{t}\right)^t} = \left(\frac{t}{t-1}\right)^t$$

$$= \left(\frac{\boxed{t-1} + 1}{\boxed{t-1}}\right)^{\boxed{t-1} + 1} = \underbrace{\left(1 + \frac{1}{\boxed{t-1}}\right)^{\boxed{t-1}}}_e \cdot \underbrace{\frac{t-1+1}{t-1}}_1 \rightarrow e$$

$$\begin{aligned} \left(\frac{n-1}{n}\right)^n &= \frac{n-1}{n-1+1} = \frac{1}{\left(\frac{n-1+1}{n-1}\right)^{n-1+1}} \\ &= \frac{1}{\underbrace{\left(1 + \frac{1}{n-1}\right)^{n-1}}_e \cdot \underbrace{\left(1 + \frac{1}{n-1}\right)}_1} \\ &\rightarrow \frac{1}{e} = e^{-1} \end{aligned}$$

常用



例1.

$$\lim_{n \rightarrow \infty} \underbrace{\left(1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n\right)}_{X_n} \quad \downarrow \text{存在.}$$

$$\ln n = \ln n - \ln(n-1) + \ln(n-1)$$

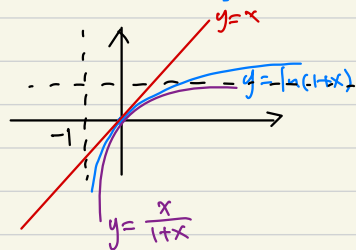
$$1^\circ. X_{n+1} - X_n = \frac{1}{n+1} - \ln(n+1) + \ln(n) = \frac{1}{n+1} - \ln\left(1 + \frac{1}{n}\right) < 0$$

$$\begin{aligned} 2^\circ. X_n &= \sum_{k=1}^n \frac{1}{k} - \sum_{k=1}^{n-1} \ln\left(1 + \frac{1}{k}\right) \\ &= \sum_{k=1}^{n-1} \left(\frac{1}{k} - \ln\left(1 + \frac{1}{k}\right) \right) + \frac{1}{n} > 0 \end{aligned}$$

X_n 单调递减且有下界0, 从而收敛.

对数不等式.

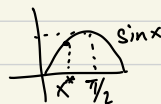
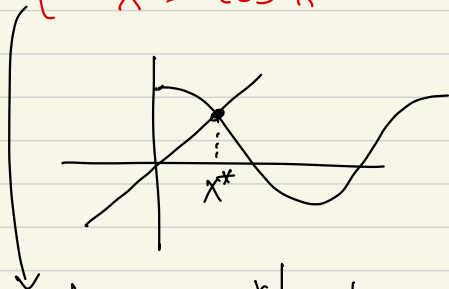
$$\frac{x}{1+x} \leq \ln(1+x) \leq x$$



$$\begin{aligned} \text{应用: } \lim_{n \rightarrow \infty} \underbrace{\left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right)}_{X_{2n} - X_n} &= \ln 2 \\ &\quad \downarrow \quad \downarrow \\ &\quad 0 \quad \ln 2 \end{aligned}$$

121.

$$\left\{ \begin{array}{l} X_{n+1} = \cos X_n \\ \quad \quad \quad \hookrightarrow X^* \\ X^* = \cos X^* \end{array} \right. , \quad X_{n+1} = \sin X_n \checkmark \hookrightarrow 0.$$



和差化积

$$2 \left| \sin \frac{X_n + X^*}{2} \right| \left| \sin \frac{X_n - X^*}{2} \right|$$

$$\leq 2 \underbrace{\left| \sin \frac{X^* + \pi/2}{2} \right|}_{\text{记为 } \rho} \left| \frac{X_n - X^*}{2} \right|$$

$$|X_{n+1} - X^*| = |\cos X_n - \cos X^*| \leq \rho \cdot |X_n - X^*|, \quad \rho < 1.$$

$$\leq \underbrace{(\rho^{n+1})}_{\rightarrow 0} |X_0 - X^*| \rightarrow 0$$

$$\lim_{n \rightarrow \infty} |X_n - X^*| = 0$$

$$\Downarrow$$

$$\lim_{n \rightarrow \infty} (X_n - X^*) = 0 \Rightarrow \lim_{n \rightarrow \infty} X_n = X^*$$

$$\left| \frac{a_n}{n} \right| \quad 0 \leq a_{m+n} \leq a_m + a_n \quad , \quad \lim_{n \rightarrow \infty} \frac{a_n}{n} \text{ 存在.}$$

$$m = qn + r \quad r \in \{0, 1, \dots, n-1\}$$

$$\frac{a_m}{m} = \frac{a_{qn+r}}{qn+r} \leq \frac{qa_n + a_r}{qn+r} \rightarrow \frac{a_n}{n} \quad , \quad \begin{matrix} m \rightarrow \infty \\ q \rightarrow \infty \end{matrix}$$

$$\limsup_{m \rightarrow \infty} \frac{a_m}{m} \leq \inf \frac{a_n}{n} \leq \liminf_{m \rightarrow \infty} \frac{a_m}{m}$$

$\uparrow$ 由定义

$$* \text{ 取 } B = \inf \left\{ \frac{a_n}{n} \right\}. \quad \frac{a_n}{n} \rightarrow B.$$

§1.8 函数的连续性与间断点

如何定义 $y(t)$ 的连续性?

技巧: 函数在一点处的连续性.

间断点 = 不连续的点.

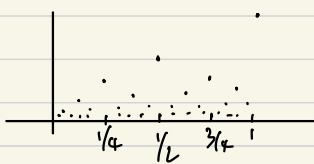
第I类 情况1: 左右极限存在
第II类 情况2: 左右极限不存在

数列定义

① 定义: $f(x)$ 在 x_0 处 连续 $\Leftrightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0)$

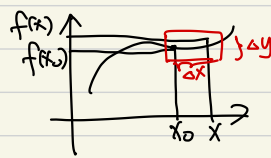
$D(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$, $D(x)$ 在 $\forall x_0$ 不连续

$R(x) = \begin{cases} \frac{1}{p}, & x = \frac{q}{p} \\ 0, & x \notin \mathbb{Q} \end{cases}$, $R(x)$ 在无理点处连续
有理点处不连续 ($\neq 0$)



$\lim_{x \rightarrow x_0} R(x) = 0$

$\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = 0$
 $\lim_{\substack{x-x_0 \rightarrow 0 \\ \Delta x}} (f(x) - f(x_0)) = 0$
 Δy



$\lim_{\Delta x \rightarrow 0} \Delta y = 0$
 $\Delta y \rightarrow 0$ if $\Delta x \rightarrow 0$

② 左连续: $\lim_{x \rightarrow x_0^-} f(x) = f(x_0)$
 $\rightarrow f(x_0^-), f(x_0-0)$

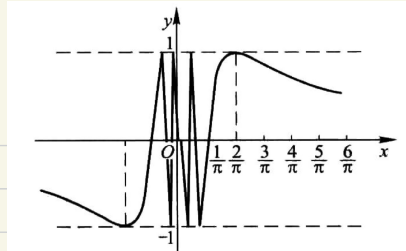
右连续: $\lim_{x \rightarrow x_0^+} f(x) = f(x_0)$
 $\rightarrow f(x_0^+), f(x_0+0)$

记号: 单侧极限

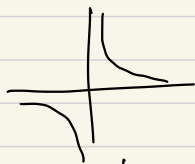
$f(x_0^+) = f(x_0^-) = f(x_0)$

例 $\sin \frac{1}{x}$, 在 $x=0$ 处. $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ 不存在.

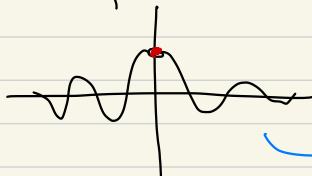
→ 振荡间断点. (II)



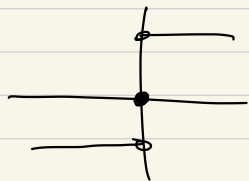
例. $\frac{1}{x}$ 在 $x=0$ 处. 无穷间断点. (II)



例. $\frac{\sin x}{x}$ 在 $x=0$ 处. 第一类间断点.
→ 可去间断点.



例. $\text{sgn}(x)$ 在 $x=0$ 处. I



$$\left. \begin{array}{l} \text{sgn}(0^-) = -1 \\ \text{sgn}(0^+) = +1 \end{array} \right\} \neq 0.$$

跳跃间断点.

⊕ 定义: 函数 $f(x)$ 在定义域 I 上连续

$f(x)$ 在 $\forall x_0 \in I$ 都连续.

↑
边界点. 考虑单侧极限.

2022.10.5

习题

(大胆假设, 小心求证)

$\sqrt[n]{a} \rightarrow 1, a > 0, n \rightarrow \infty$

$\lim_{n \rightarrow +\infty} (1 - \frac{1}{n})^{n^2} = ? \quad 0$

$((1 - \frac{1}{n})^n)^n \rightarrow (\frac{1}{e})^n \rightarrow 0$

Tip: 夹逼准则
或取对数.

$\lim_{n \rightarrow +\infty} (1 - \frac{1}{n^2})^n = ? \quad 1$

$\sqrt[n]{(1 - \frac{1}{n^2})^{n^2}} \rightarrow \sqrt[n]{\frac{1}{e}} \rightarrow 1$

$\lim_{n \rightarrow +\infty} \frac{e^n}{(1 + \frac{1}{n})^{n^2}} = ? \quad \sqrt{e}$

$(\frac{e}{(1 + \frac{1}{n})^n})^n \rightarrow (\frac{e}{e})^n \rightarrow 1$

$\lim_{n \rightarrow +\infty} (1 + \frac{1}{2!} + \dots + \frac{1}{n!}) = ? \quad e$

$x_n = (1 + \frac{1}{n})^n = 1 + 1 + \frac{1}{2!}(1 - \frac{1}{n}) + \dots + \frac{1}{n!}(1 - \frac{1}{n}) \dots (1 - \frac{n-1}{n})$

$1 + 1 + \frac{1}{2!}(1 - \frac{1}{n}) + \dots + \frac{1}{k!}(1 - \frac{1}{n}) \dots (1 - \frac{k-1}{n}) \leq x_n \leq 1 + 1 + \frac{1}{2!} + \dots + \frac{1}{n!}$

$\lim_{k \rightarrow \infty} y_k \leq e \leq \lim_{n \rightarrow \infty} y_n$

Tip: $q^n \rightarrow 1, q > 1$

$\sqrt[n]{1+a} - 1 < \frac{a}{n}$

$1+a < (1 + \frac{a}{n})^n$

$\frac{\sin 6x + x f(x)}{x^3} \rightarrow 0, \frac{6 + f(x)}{x^2} \rightarrow ?$

$\sim \frac{6x + x f(x)}{x^3} = \frac{6 + f(x)}{x^2} \rightarrow 0$

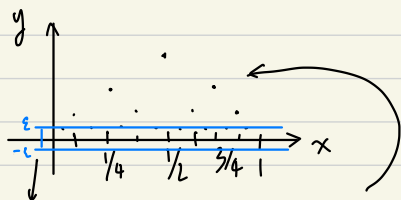
$\sin 6x = 6x - \frac{1}{3!}(6x)^3 + o(x^3)$

$\frac{\sin 6x + x f(x)}{x^3} = \frac{6 + x f(x)}{x^2} - 36 + \frac{o(x^3)}{x^3}$

小心求证: $\sin(6x) = 6x + o(6x), \frac{6x + x f(x) + o(6x)}{x^3} = \frac{6 + x f(x)}{x^2} + \frac{o(6x)}{x^3}$

$\lim_{x \rightarrow x_0} R(x) = 0$ 的证明思路

$$R(x) = \begin{cases} \frac{1}{p}, & x = \frac{q}{p} \\ 0, & x \notin \mathbb{Q} \end{cases}, x \in (0,1)$$



重点: $\forall \varepsilon > 0$, 带子外面只有有限个点, 记为 $x_1, x_2, \dots, x_k$

于是可取 $\delta = \min \{ |x_k - x_0| \mid x_k \neq x_0, k=1, \dots, k \}$

显然 $\delta > 0$, 并且 $0 < |x - x_0| < \delta$ 时 $|R(x)| < \varepsilon$

故 $\lim_{x \rightarrow x_0} R(x) = 0$.

相关知识

用有理数 $\frac{q}{p}$ 去近似无理数时, p 要比较大才行!

例: $\pi \approx \frac{22}{7}, \frac{335}{113}, \dots$

§ 1.8 函数的连续性与间断点

如何定义 $y(t)$ 的连续性?

技巧: 函数在一点处的连续性.

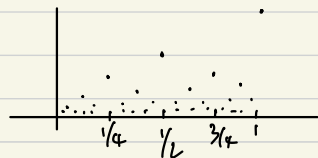
间断点 = 不连续的点.

第I类 情况1: 左右极限存在
第II类 情况2: 左右极限不存在

① 定义: $f(x)$ 在 x_0 处 连续 $\Leftrightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0)$

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}, \quad D(x) \text{ 在 } \forall x_0 \text{ 不连续}$$

$$R(x) = \begin{cases} \frac{1}{p}, & x = \frac{q}{p} \\ 0, & x \notin \mathbb{Q} \end{cases}, \quad R(x) \text{ 在无理点处连续, 有理点处不连续.}$$



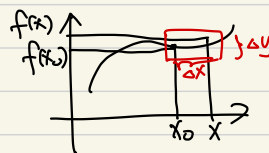
$$\lim_{x \rightarrow x_0} R(x) = 0$$

注: $0 = \frac{0}{1}$
 $R(0) = 1$

$$\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = 0$$

$$\lim_{\substack{x-x_0 \rightarrow 0 \\ \Delta x}} (f(x) - f(x_0)) = 0$$

Δx Δy



$$\lim_{\Delta x \rightarrow 0} \Delta y = 0$$

$\Delta y \rightarrow 0 \text{ if } \Delta x \rightarrow 0$

② 左连续: $\lim_{x \rightarrow x_0^-} f(x) = f(x_0)$

右连续: $\lim_{x \rightarrow x_0^+} f(x) = f(x_0)$

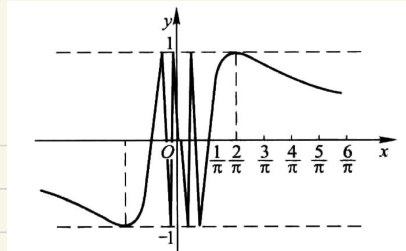
$\hookrightarrow f(x_0^-), f(x_0+0)$

记号: 单侧极限

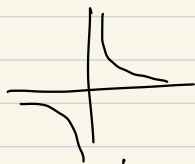
$$f(x_0^+) = f(x_0^-) = f(x_0)$$

例 $\sin \frac{1}{x}$, 在 $x=0$ 处. $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ 不存在.

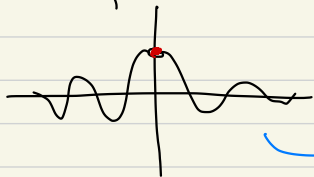
→ 振荡间断点. (II)



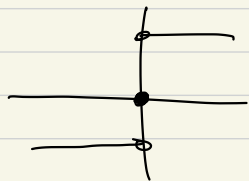
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例. $\frac{\sin x}{x}$ 在 $x=0$ 处. 第一类间断点.
→ 可去间断点.



例. $\text{sgn}(x)$ 在 $x=0$ 处. I



$$\left. \begin{array}{l} \text{sgn}(0^-) = -1 \\ \text{sgn}(0^+) = +1 \end{array} \right\} \neq 0.$$

跳跃间断点.

⊕ 定义: 函数 $f(x)$ 在定义域 I 上连续

$f(x)$ 在 $\forall x_0 \in I$ 都连续.

↑
边界点. 考虑单侧极限.

2022.10.5

习题

(大胆假设, 小心求证)

$$\sqrt[n]{a} \rightarrow 1, \quad a > 0, \quad n \rightarrow \infty$$

$$\lim_{n \rightarrow +\infty} (1 - \frac{1}{n})^{n^2} = ? \quad 0$$

$$((1 - \frac{1}{n})^n)^n \rightarrow (\frac{1}{e})^n \rightarrow 0 \quad ?$$

Tip: 夹逼原理
或取对数.

$$\lim_{n \rightarrow +\infty} (1 - \frac{1}{n^2})^n = ? \quad 1$$

$$\sqrt[n]{(1 - \frac{1}{n^2})^{n^2}} \rightarrow \sqrt[n]{\frac{1}{e}} \rightarrow 1 \quad ?$$

$$\lim_{n \rightarrow +\infty} \frac{e^n}{(1 + \frac{1}{n})^{n^2}} = ? \quad \sqrt{e}$$

$$(\frac{e}{(1 + \frac{1}{n})^n})^n \rightarrow (\frac{e}{e})^n \rightarrow 1 \quad \times$$

$$\lim_{n \rightarrow +\infty} (1 + \frac{1}{2!} + \dots + \frac{1}{n!}) = ? \quad e.$$

$$x_n = (1 + \frac{1}{n})^n = 1 + 1 + \frac{1}{2!}(1 - \frac{1}{n}) + \dots + \frac{1}{n!}(1 - \frac{1}{n}) \dots (1 - \frac{n-1}{n})$$

$$1 + 1 + \frac{1}{2!}(1 - \frac{1}{n}) + \dots + \frac{1}{k!}(1 - \frac{1}{n}) \dots (1 - \frac{k-1}{n}) \leq x_n \leq 1 + 1 + \frac{1}{2!} + \dots + \frac{1}{n!}$$

$$\begin{aligned} & \downarrow \quad \downarrow \quad \downarrow \\ & y_k \leq e \leq \lim_{n \rightarrow \infty} y_n \\ & \lim_{k \rightarrow \infty} y_k \leq e \leq \lim_{n \rightarrow \infty} y_n \end{aligned}$$

Tip:

$$q^n \rightarrow 1, \quad q > 1$$

$$\sqrt[n]{1+a} - 1 < \frac{a}{n}$$

$$1+a < (1 + \frac{a}{n})^n$$

$$\frac{\sin 6x + x f(x)}{x^3} \rightarrow 0, \quad \frac{6 + f(x)}{x^2} \rightarrow ?$$

$$\sim \frac{6x + x f(x)}{x^3} = \frac{6 + f(x)}{x^2} \rightarrow 0$$

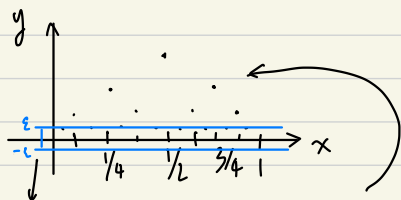
$$\sin 6x = 6x - \frac{1}{3!}(6x)^3 + o(x^3)$$

$$\frac{\sin(6x) + x f(x)}{x^3} = \frac{6 + x f(x)}{x^2} - 36 + \frac{o(x^3)}{x^3}$$

$$\text{小心求证: } \sin(6x) = 6x + o(6x), \quad \frac{6x + x f(x) + o(6x)}{x^3} = \frac{6 + x f(x)}{x^2} + \frac{o(6x)}{x^3} \rightarrow ?$$

$\lim_{x \rightarrow x_0} R(x) = 0$ 的证明思路

$$R(x) = \begin{cases} \frac{1}{p}, & x = \frac{q}{p} \\ 0, & x \notin \mathbb{Q} \end{cases}, x \in (0,1)$$



重点: $\forall \varepsilon > 0$, 带子外面只有有限个点, 记为 $x_1, x_2, \dots, x_k$

于是可取 $\delta = \min \{ |x_k - x_0| \mid x_k \neq x_0, k=1, \dots, k \}$

显然 $\delta > 0$, 并且 $0 < |x - x_0| < \delta$ 时 $|R(x)| < \varepsilon$

故 $\lim_{x \rightarrow x_0} R(x) = 0$.

相关知识

用有理数 $\frac{q}{p}$ 去近似无理数时, p 要比较大才行!

例: $\pi \approx \frac{22}{7}, \frac{335}{113}, \dots$

§1.9. 连续函数的运算与初等函数的连续性

★ “初等函数都是连续的”

↓

证明: $+-\times\div$

定义域: 自然定义域.

定理: 四则运算, $f(x), g(x)$ 在 x_0 连续.

page 62

$\Rightarrow f \pm g, fg, f/g (g \neq 0)$ 在 x_0 连续.

复合运算. $g(x)$ 在 x_0 连续, $f(x)$ 在 $u_0 = g(x_0)$ 连续

$\Rightarrow f(g(x))$ 在 x_0 连续

★ Tip: 若 $\lim_{x \rightarrow x_0} g(x) = u_0$, 且 $f(u)$ 在 u_0 连续. $\Rightarrow \lim_{x \rightarrow x_0} f(g(x)) = \lim_{u \rightarrow u_0} f(u) = f(u_0)$

(不要求 $g(x)$ 在 x_0 连续)

$$\lim_{x \rightarrow x_0} f(g(x)) = f\left(\lim_{x \rightarrow x_0} g(x)\right)$$

↑
连续. $\lim$ 和求值可交换顺序.

proof.

反函数

$f(x)$ 严格单调, 且连续

$\Rightarrow f^{-1}(x)$ 也连续. page 62

$$|x| = \sqrt{x^2}$$

$$\operatorname{sgn}(x) = \frac{x}{|x|}, (x \neq 0 \text{ 时})$$

↓
自然定义域: $(-\infty, 0) \cup (0, +\infty)$

⊗ 基本初等函数都连续

$$x^u, \boxed{a^x}, \boxed{\sin(x)}, \cos(x), \dots, \log_a(x).$$

$\sin(x) = \sin(x + \frac{\pi}{2})$

$\parallel$
 $e^{\ln x^u} = e^{u \ln x} \rightarrow a^x = e^{\ln a^x} = \boxed{e^x}^{\ln a}$
 $\swarrow$
 $\arcsin(x), \arccos(x), \dots$

1°. $\sin(x)$ 在 $\forall x_0$ 连续. $\lim_{x \rightarrow x_0} \sin(x) = \sin(x_0)$

$$|\sin(x) - \sin(x_0)| = 2 \left| \sin \frac{x-x_0}{2} \cos \frac{x+x_0}{2} \right| \leq |x-x_0| < \varepsilon.$$

2°. e^x 在 $\forall x_0$ 连续. $\lim_{x \rightarrow x_0} e^x = e^{x_0} \Leftrightarrow \lim_{x \rightarrow x_0} e^{x-x_0} = 1$

直接证明

$$\begin{cases} e^x - 1 < \varepsilon, & x > 0 \\ 1 < e^x \leq 1 + \varepsilon \\ 0 < x < \ln(1+\varepsilon) \\ \text{取 } \delta = \ln(1+\varepsilon) \end{cases}$$

常用
技巧

把 x 取为 $\frac{1}{n}$

$$\lim_{x \rightarrow 0^+} e^x = 1 \quad e^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{e} = 1$$

$$0 < \sqrt[n]{e} - 1 < \varepsilon \Leftrightarrow e < (1+\varepsilon)^n, \quad (1+\varepsilon)^n > 1+n\varepsilon$$

$\uparrow$

$$\text{取 } N = \left[\frac{e-1}{\varepsilon} \right] + 1, \quad n > N.$$

$$\begin{aligned} e &< 1+n\varepsilon < (1+\varepsilon)^n \\ \uparrow \\ n &> \frac{e-1}{\varepsilon} \end{aligned}$$

$$= 1 + n\varepsilon + \frac{n(n-1)}{2} \varepsilon^2 + \dots + \varepsilon^n$$

2022.10.7

2.14

$a > 0$

$$\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \lim_{x \rightarrow 0} \log_a \left[(1+x)^{\frac{1}{x}} \right] = \log_a e = \frac{1}{\ln a}$$

技巧:
无穷小视为整体

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{x \rightarrow 0} \frac{t}{\log_a(1+t)} = \frac{1}{\frac{1}{\ln a}} = \ln a$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^2 - 1}{x} = \lim_{x \rightarrow 0} \frac{t}{e^{\frac{1}{2} \ln(1+t)} - 1} = \lim_{t \rightarrow 0} \frac{\frac{1}{2} \ln(1+t)}{e^{\frac{1}{2} \ln(1+t)} - 1} \cdot \frac{t}{\frac{1}{2} \ln(1+t)} = 2$$

$(1+x)^2 = 1+t$
 $2 \ln(1+x) = \ln(1+t)$
 $x = e^{\frac{1}{2} \ln(1+t)} - 1$

$\frac{6x}{\sin x} \rightarrow 6$
 $\frac{\ln(1+2x)}{2x} \rightarrow 1$

$$\lim_{x \rightarrow 0} (1 + \frac{2x}{\sin x})^{\frac{3}{\sin x}} = \lim_{\substack{t \rightarrow 0 \\ x \rightarrow 0}} (1 + \frac{1}{t})^t \sim e^6$$

$x = e^{\ln x}$

$$(1 + \frac{1}{x})^x$$

技巧: $u(x)^{v(x)} = e^{\ln u(x) \cdot v(x)} = e^{v(x) \ln u(x)}$

2022.10.7

习题

$a, b, c > 0$

★ $\lim_{n \rightarrow \infty} \left(\frac{\sqrt[n]{a} + \sqrt[n]{b} + \sqrt[n]{c}}{3} \right)^n = ?$

$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{a^n + b^n + c^n}{3}} = ?$

$\lim_{k \rightarrow \infty} \left[\lim_{n \rightarrow \infty} (\cos(k! \pi x))^{2^n} \right] = ?$

$\sqrt[n]{a} = 1 + \frac{1}{n} \ln a + \left(\frac{1}{n} o(1) \right) \circ \left(\frac{1}{n} \right)$

$n(\sqrt[n]{a} - 1) - \ln a = o(\ln a)$

$\left(\frac{1 + \frac{\ln a}{n} + 1 + \frac{\ln b}{n} + 1 + \frac{\ln c}{n} + o\left(\frac{1}{n}\right)}{3} \right)^n$

$\left(1 + \frac{\ln(abc)^{1/3} + o(1)}{n} \right)^n \rightarrow e^{\ln(abc)^{1/3}} = (abc)^{1/3}$

$\left(1 + \frac{A}{n} \right)^n = \left(1 + \frac{A}{n} \right)^{n/A \cdot A} \rightarrow e^A$

§1.4 $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$, $\Downarrow$ $x = \frac{1}{n}$

$\sqrt[3]{abc}$

$n(\sqrt[n]{a} - 1) \rightarrow \ln a$

$\max(a, b, c)$

夹逼原理

$D(x)$

$f(x) \sim g(x)$

$f(x) - g(x) \rightarrow 0$

$f(x) - g(x) = o(g(x))$

2022.10.7
Fri

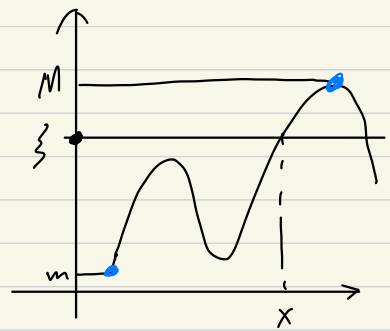
§1.10 闭区间上的连续函数.

非常特别

$$f(x) \in C(\Omega)$$

$$[0,1], (0,1)$$

$$\text{记号: } f(x) \in C([0,1])$$



①. 有最值 (最大, 最小值)
 M, m

②. 介值定理. $\forall \xi \in [m, M], \exists x. \text{ s.t. } f(x) = \xi$

" ξ 取0" 零点定理. $g(x) = f(x) - \xi$

③. 值域也是闭区间.

④. 一致连续.

$$\text{定义: } \forall \varepsilon > 0. \exists \delta > 0. \text{ s.t. } |f(x_1) - f(x_2)| < \varepsilon, \text{ if } |x_1 - x_2| < \delta.$$

连续 vs 一致连续. $\rightarrow \delta$ 可以与 x_0 没有关系.

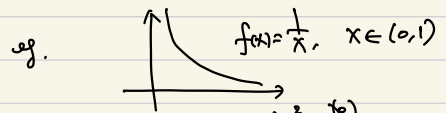
eg: $\sqrt{x}$, 在 $[0,1]$ 上一致连续

1° 每点都连续

$\rightarrow$ 与 x_0, ε 都有关系.

$$2^\circ. x_0 \text{ 处. } \forall \varepsilon > 0. \exists \delta > 0. \text{ s.t. } |f(x) - f(x_0)| < \varepsilon, |x - x_0| < \delta.$$

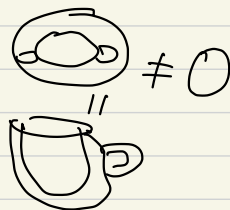
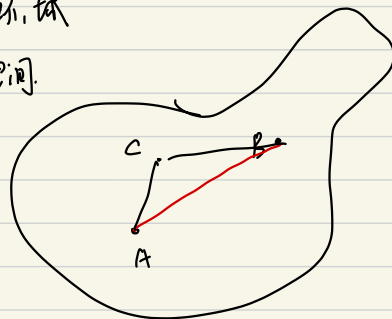
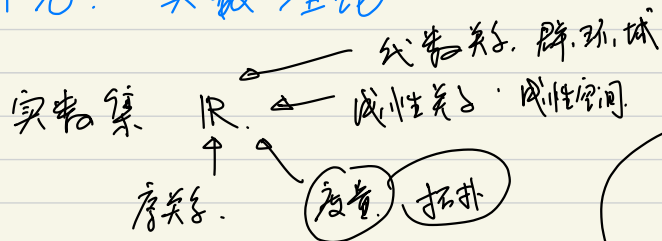
注: $\frac{1}{x}$ 在 $(0,1)$ 上不是 一致连续 的.



$$= \left| \frac{x - x_0}{x x_0} \right|$$

$$\forall \varepsilon > 0. \text{ 取 } \delta = \min\left(x_0^2 \varepsilon, \frac{x_0}{2}\right). \text{ s.t. } \left| \frac{1}{x} - \frac{1}{x_0} \right| < \varepsilon, |x - x_0| < \delta.$$

补充：实数理论



④ 实数的基本理论

1° 完备性/连续性. $x \leq y$, $\exists c, x \leq c \leq y$

2° 确界原理. 有上下界, 则必有上下确界 (唯一) $\xrightarrow{\text{应用}}$ 单调有界数列有极限.

3° 阿基米德原理. $h > 0, \forall x \in \mathbb{R}, \exists n, (n-1)h \leq x < nh$.

4° 闭区间套原理 (柯西-康托尔原理).

$I_n \subset I_{n-1}, \exists c, s.t. c \in I_n, \forall n$.

5° 有限覆盖原理 (博雷尔-勒贝格原理).

开区间 $(0, \frac{1}{n})$ 不行!

6° 聚点定理 (波尔查诺-魏尔斯特拉斯原理).

闭区间. 用开区间盖住. 开区间 $(0, 1)$. $(\frac{x}{2}, 1)$ 不行!

$\hookrightarrow \{x_n\}$ 有界, 则存在 $\{x_{n_k}\}$ 收敛.

x_1, x_2, x_3

$(0, 1) \subset \bigcup_{k \in \mathbb{N}} (\frac{x_k}{2}, 1)$

$a \in (\frac{x}{2}, 1)$

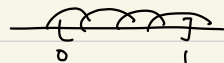
7° Cauchy 准则. $|x_n - x_m| < \varepsilon$.

$\mathbb{K}$
 $\{a_1, \dots\}$

④ 应用. 闭区间上的连续函数 $f(x)$, $I = [0, 1]$ | $I = [a, b]$ 同理.

1°. 有最大/小值. (魏尔斯特拉斯最大/小值定理)

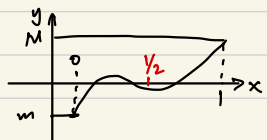
证明: i). 有界, $\Leftarrow$ 局部有界 + 有限(开)覆盖引理.



ii). 确界可达 $\Leftarrow$ 上确界记为 M . 考虑 $g(x) = \frac{1}{M - f(x)}$ 的连续性和有界性
不可在 I 上连续 (反证法)

2°. 零点定理 (波尔查诺-柯西中值定理)

证明: 若 $M > 0$, $m < 0$, 则 $\exists x$ s.t. $f(x) = 0$.



$$a_0 = 0, \quad b_0 = 1, \quad I_0 = [a_0, b_0]$$

$$f\left(\frac{a_0 + b_0}{2}\right) \geq 0?$$

$$I_1 = [a_1, b_1]$$

$$I_n \subset I_{n-1}$$

$$I_n = [a_n, b_n], \quad (b_n - a_n) \rightarrow 0$$

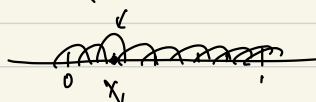
闭区间套引理 $\Rightarrow c \in I_n, f(c) = 0$.

3°. 一致连续性

$$|f(x_1) - f(x_2)| < \varepsilon, \quad |x_1 - x_2| < \delta.$$

证明:

$$(x_i - \delta_i, x_i + \delta_i)$$

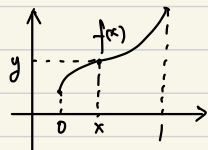


$$[0, 1] \subset \bigcup_{x_i} (x_i - \delta_i, x_i + \delta_i)$$

有限覆盖引理, 取 $\rho = \min \{\delta_i^{(k)}\}$

40. 若 $f(x)$ 严格单调, 则 $f^{-1}(x)$ 也是连续函数.

证明:



对 $f^{-1}(x)$ 使用这个引理.

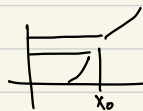
$$f^{-1}(J) = I$$

引理: 若 f 在 $I = [a, b]$ 上单调增

$$f \in C([a, b]) \iff f(I) = [f(a), f(b)] = J$$

$\Rightarrow$ 介值定理

$\Leftarrow$ 反证法



补充: 上/下极限.

压缩映射

$$|f(x) - f(y)| \leq L|x - y|.$$

$$0 \leq L < 1$$

Lipschitz 连续